MH4110 PDE

Tutorial 07

Question 1

Solve the inhomogeneous heat equation on the half-line with Dirichlet boundary condition

$$\begin{aligned} &u_t - ku_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &u(0,t) = 0, \quad t \geq 0, \\ &u(x,0) = \phi(x) \end{aligned}$$

using the method of reflection.

Question 1 (Solution):

We employ the method of reflection to reduce the problem to one on the whole real axis. Take the odd extensions:

$$F(x,t) = \begin{cases} f(x,t), & x > 0, \\ -f(-x,t), & x < 0, \\ 0, & x = 0, \end{cases} \quad \Phi(x) = \begin{cases} \phi(x), & x > 0, \\ -\phi(-x), & x < 0, \\ 0, & x = 0, \end{cases} \tag{1}$$

Then we solve the extended heat equation on the whole real axis:

$$\begin{aligned} &U_t - kU_{xx} = F(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &U(x,0) = \Phi(x), \quad -\infty < x < \infty, \end{aligned}$$

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$\begin{aligned} U(x,t) = &\frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \Phi(y)dy \\ &+ \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] F(y,s)dyds \end{aligned}$$

Question 1 (Solution):

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$
\begin{aligned}
U(x,t) &= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \Phi(y)dy \\
&\quad + \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] F(y,s)dyds \\
&= -\frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{0} \exp\left[-\frac{(x-y)^2}{4kt}\right] \phi(-y)dy \\
&\quad + \frac{1}{\sqrt{4\pi kt}} \int_{0}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \phi(y)dy \\
&\quad - \int_0^t \int_{-\infty}^{0} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] f(-y,s)dyds \\
&\quad + \int_0^t \int_{0}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] f(y,s)dyds
\end{aligned}
$$

use $-y$ to replace y

Question 1 (Solution):

Making the change of variables in the first and third integrals as $y \to -y$ gives

$$
\begin{aligned}
&U(x,t) \\
&= \frac{1}{\sqrt{4\pi kt}} \int_{\infty}^{0} \exp\left[-\frac{(x+y)^2}{4kt}\right] \phi(y)dy \\
&\quad + \frac{1}{\sqrt{4\pi kt}} \int_{0}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \phi(y)dy \\
&\quad + \int_0^t \int_{\infty}^{0} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right] f(y,s)dyds \\
&\quad + \int_0^t \int_{0}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] f(y,s)dyds \\
&= \frac{1}{\sqrt{4\pi kt}} \int_{0}^{\infty} \left\{\exp\left[-\frac{(x-y)^2}{4kt}\right] - \exp\left[-\frac{(x+y)^2}{4kt}\right]\right\} \phi(y)dy \\
&\quad + \int_0^t \int_{0}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \left\{\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]\right\} f(y,s)dyds.
\end{aligned}
$$

Question 1 (Solution):

We know that $u(x,t) = U(x,t)$ on the positive axis ($x > 0$). So, the solution $u(x,t)$ ($x > 0$) can be expressed as

$$
\begin{aligned}
&u(x,t) \\
&= \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \left\{ \exp\left[-\frac{(x-y)^2}{4kt}\right] - \exp\left[-\frac{(x+y)^2}{4kt}\right] \right\} \phi(y)dy \\
&\quad + \int_0^t \int_0^\infty \frac{1}{\sqrt{4\pi k(t-s)}} \left\{ \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right] \right\} f(y,s)dyds.
\end{aligned}
$$

Question 2

Solve the Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_t - kv_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = \underline{h(t)} \quad \text{(boundary condition at } x = 0\text{)}, \\ &v(x,0) = \phi(x) \quad \text{(initial condition at } t = 0\text{)}. \end{aligned}$$

Question 2 (Solution):

We employ the subtraction method to reduce the problem to one with zero Dirichlet data, and then use the reflection method to derive a solution formula for the reduced problem. Let

$$\boxed{V(x,t) = v(x,t) - h(t).}$$

$V(0,t)$
$= v(0,t) - h(t)$
$= h(t) - h(t)$
$= 0$

We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

$$\begin{cases} V_t - kV_{xx} = f(x,t) - h'(t), \quad 0 < x < \infty, \quad t > 0, \\ V(0,t) = 0, \\ V(x,0) = \phi(x) - h(0). \end{cases}$$

This new problem has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions. For simplicity, let $F(x,t) = f(x,t) - h'(t)$ and $\Phi(x) = \phi(x) - h(0)$, we have

$$\begin{aligned} &V_t - kV_{xx} = F(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0, \\ &V(x,0) = \Phi(x). \end{aligned}$$

Question 2 (Solution):

Take the odd extensions:

$$\tilde{F}(x,t) = \begin{cases} F(x,t), & x > 0, \\ -F(-x,t), & x < 0, \\ 0, & x = 0, \end{cases} \quad \tilde{\Phi}(x) = \begin{cases} \Phi(x), & x > 0, \\ -\Phi(-x), & x < 0, \\ 0, & x = 0, \end{cases} \tag{2}$$

Then we solve the extended heat equation on the whole real axis:

$$\begin{aligned} &U_t - kU_{xx} = \tilde{F}(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &U(x,0) = \tilde{\Phi}(x), \quad -\infty < x < \infty, \end{aligned}$$

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$\begin{aligned} U(x,t) = &\frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \tilde{\Phi}(y)dy \\ &+ \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] \tilde{F}(y,s)dyds. \end{aligned}$$

Question 2 (Solution):

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$
\begin{aligned}
U(x,t) &= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \tilde{\Phi}(y)dy \\
&\quad + \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] \tilde{F}(y,s)dyds \\
&= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{0} \exp\left[-\frac{(x-y)^2}{4kt}\right] \tilde{\Phi}(y)dy \\
&\quad + \frac{1}{\sqrt{4\pi kt}} \int_{0}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \tilde{\Phi}(y)dy \\
&\quad + \int_0^t \int_{-\infty}^{0} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] \tilde{F}(y,s)dyds \\
&\quad + \int_0^t \int_{0}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] \tilde{F}(y,s)dyds
\end{aligned}
$$

$$
\begin{aligned}
U(x,t) = &-\frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{0}\exp\left[-\frac{(x-y)^2}{4kt}\right]\Phi(-y)dy \\
&+\frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x-y)^2}{4kt}\right]\Phi(y)dy \\
&-\int_0^t\int_{-\infty}^{0}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]F(-y,s)dyds \\
&+\int_0^t\int_{0}^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]F(y,s)dyds
\end{aligned}
$$

Making the change of variables in the first and third integrals as $y=-y$ gives

$$
\begin{aligned}
&U(x,t) \\
&= -\frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x+y)^2}{4kt}\right]\Phi(y)dy + \frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x-y)^2}{4kt}\right]\Phi(y)dy \\
&\quad -\int_0^t\int_{0}^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]F(y,s)dyds \\
&\quad +\int_0^t\int_{0}^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]F(y,s)dyds
\end{aligned}
$$

Question 2 (Solution):

$$
\begin{aligned}
&U(x,t)\\
&= \frac{1}{\sqrt{4\pi kt}}\int_0^\infty \left\{\exp\left[-\frac{(x-y)^2}{4kt}\right] - \exp\left[-\frac{(x+y)^2}{4kt}\right]\right\}\Phi(y)dy\\
&\quad + \int_0^t\int_0^\infty \frac{1}{\sqrt{4\pi k(t-s)}}\left\{\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]\right\}F(y,s)dyds.
\end{aligned}
$$

We know that $V(x,t) = U(x,t)$ on the positive axis ($x > 0$). So, the solution $v(x,t) = V(x,t) + h(t)$ ($x > 0$) can be expressed as

$$
\begin{aligned}
&v(x,t)\\
&= h(t) + \frac{1}{\sqrt{4\pi kt}}\int_0^\infty \left\{\exp\left[-\frac{(x-y)^2}{4kt}\right] - \exp\left[-\frac{(x+y)^2}{4kt}\right]\right\}\Phi(y)dy\\
&\quad + \int_0^t\int_0^\infty \frac{1}{\sqrt{4\pi k(t-s)}}\left\{\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]\right\}F(y,s)dyds
\end{aligned}
$$

Substituting $\Phi(x) = \phi(x) - h(0)$ and $F(x,t) = f(x,t) - h'(t)$ into the above expression gives the explicit solution formula for this Dirichlet problem.

Question 2 (Solution):

Substituting $\Phi(x) = \phi(x) - h(0)$ and $F(x,t) = f(x,t) - h'(t)$ into the above expression gives the explicit solution formula for this Dirichlet problem of a boundary source $h(t)$ on the half-line as

$$
\begin{aligned}
&v(x,t) \\
&= h(t) + \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \left\{ \exp\left[-\frac{(x-y)^2}{4kt}\right] - \exp\left[-\frac{(x+y)^2}{4kt}\right] \right\} [\phi(y) - h(0)]\, dy \\
&\quad + \int_0^t \int_0^\infty \frac{1}{\sqrt{4\pi k(t-s)}} \left\{ \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right] \right\} \left[f(y,s) - h'(s)\right] dyds.
\end{aligned}
$$

In terms of the heat kernel $S(x,t)$, the solution can be rewritten as

$$
\begin{aligned}
v(x,t) = h(t) &+ \int_0^\infty [S(x-y,t) - S(x+y,t)]\, [\phi(y) - h(0)]\, dy \\
&+ \int_0^t \int_0^\infty [S(x-y,t-s) - S(x+y,t-s)]\, [f(y,s) - h'(s)]\, dyds.
\end{aligned}
$$

Question 3

Solve the inhomogeneous diffusion equation on the half-line with Neumann boundary condition

$$
\begin{aligned}
&u_t - ku_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\
&u_x(0,t) = 0, \\
&u(x,0) = \phi(x)
\end{aligned}
$$

using the method of reflection.

Question 3 (Solution):

We employ the method of reflection to reduce the problem to one on the whole real axis. Take the even extensions:

$$F(x,t) = \begin{cases} f(x,t), & x > 0, \\ f(-x,t), & x < 0, \\ 0, & x = 0, \end{cases} \quad \Phi(x) = \begin{cases} \phi(x), & x > 0, \\ \phi(-x), & x < 0, \\ 0, & x = 0, \end{cases} \tag{3}$$

Then we solve the extended heat equation on the whole real axis:

$$\begin{aligned} &U_t - kU_{xx} = F(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &U(x,0) = \Phi(x), \quad -\infty < x < \infty, \end{aligned}$$

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$\begin{aligned} U(x,t) = &\frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-y)^2}{4kt}\right] \Phi(y)dy \\ &+ \int_0^t \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi k(t-s)}} \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] F(y,s)dyds \end{aligned}$$

Question 3 (Solution):

Using the solution formula, we have the solution to the extended heat equation on the whole real axis; that is

$$
\begin{aligned}
&U(x,t)\\
&= \frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{0}\exp\left[-\frac{(x-y)^2}{4kt}\right]\phi(-y)dy + \frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x-y)^2}{4kt}\right]\phi(y)dy\\
&\quad + \int_0^t\int_{-\infty}^{0}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]f(-y,s)dyds\\
&\quad + \int_0^t\int_{0}^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]f(y,s)dyds
\end{aligned}
$$

Making the change of variables in the first and third integrals as $y \to -y$ gives

$$
\begin{aligned}
U(x,t) = &-\frac{1}{\sqrt{4\pi kt}}\int_{\infty}^{0}\exp\left[-\frac{(x+y)^2}{4kt}\right]\phi(y)dy + \frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x-y)^2}{4kt}\right]\phi(y)dy\\
&- \int_0^t\int_{\infty}^{0}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]f(y,s)dyds\\
&+ \int_0^t\int_{0}^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]f(y,s)dyds.
\end{aligned}
$$

Question 3 (Solution):

Making the change of variables in the first and third integrals as $y \to -y$ gives

$$
\begin{aligned}
U(x,t) = &-\frac{1}{\sqrt{4\pi kt}}\int_{\infty}^{0}\exp\left[-\frac{(x+y)^2}{4kt}\right]\phi(y)dy + \frac{1}{\sqrt{4\pi kt}}\int_{0}^{\infty}\exp\left[-\frac{(x-y)^2}{4kt}\right]\phi(y)dy \\
&-\int_0^t\int_{\infty}^{0}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]f(y,s)dyds \\
&+\int_0^t\int_0^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]f(y,s)dyds \\
= &\frac{1}{\sqrt{4\pi kt}}\int_0^{\infty}\left\{\exp\left[-\frac{(x-y)^2}{4kt}\right]+\exp\left[-\frac{(x+y)^2}{4kt}\right]\right\}\phi(y)dy \\
&+\int_0^t\int_0^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\left\{\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]+\exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]\right\}f(y,s)dyds.
\end{aligned}
$$

We know that $u(x,t) = U(x,t)$ on the positive axis ($x > 0$). So, the solution $u(x,t)$ ($x > 0$) can be expressed as

$$
\begin{aligned}
u(x,t) = &\frac{1}{\sqrt{4\pi kt}}\int_0^{\infty}\left\{\exp\left[-\frac{(x-y)^2}{4kt}\right]+\exp\left[-\frac{(x+y)^2}{4kt}\right]\right\}\phi(y)dy \\
&+\int_0^t\int_0^{\infty}\frac{1}{\sqrt{4\pi k(t-s)}}\left\{\exp\left[-\frac{(x-y)^2}{4k(t-s)}\right]+\exp\left[-\frac{(x+y)^2}{4k(t-s)}\right]\right\}f(y,s)dyds.
\end{aligned}
$$

Question 4

Solve the Neumann problem of a boundary source $g(t)$ on the half-line for the heat equation

$$
\begin{aligned}
&w_t - kw_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\
&w_x(0,t) = \underline{g(t)} \quad \text{(boundary condition at } x = 0\text{)}, \\
&w(x,0) = \phi(x) \quad \text{(initial condition at } t = 0\text{)}.
\end{aligned}
$$

Question 4 (Solution):

The explicit solution formula for this Neumann problem of a boundary source $h(t)$ on the half-line as

$$
\begin{aligned}
&w(x,t) \\
&= xg(t) + \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \left\{ \exp\left[-\frac{(x-y)^2}{4kt}\right] + \exp\left[-\frac{(x+y)^2}{4kt}\right] \right\} [\phi(y) - yg(0)]\, dy \\
&\quad + \int_0^t \int_0^\infty \frac{1}{\sqrt{4\pi k(t-s)}} \left\{ \exp\left[-\frac{(x-y)^2}{4k(t-s)}\right] + \exp\left[-\frac{(x+y)^2}{4k(t-s)}\right] \right\} \left[f(y,s) - yg'(s)\right] dyds.
\end{aligned}
$$

In terms of the heat kernel $S(x,t)$, the solution can be rewritten as

$$
\begin{aligned}
w(x,t) = xg(t) &+ \int_0^\infty [S(x-y,t) + S(x+y,t)]\, [\phi(y) - yg(0)]\, dy \\
&+ \int_0^t \int_0^\infty [S(x-y,t-s) + S(x+y,t-s)] \left[f(y,s) - yg'(s)\right] dyds.
\end{aligned}
$$

Question 5

Solve $u_{tt} = c^2 u_{xx} + xt,\ u(x,0) = 0,\ u_t(x,0) = 0.$

Question 5 (Solution):

From the lecture, we have the solution formula for the wave equation with a source on the whole real axis as

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

For this problem, we have $\phi(x) = \psi(x) = 0$ and $f(x,t) = xt$. The solution is therefore

$$u(x,t) = \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} ysdyds = \frac{1}{2c}\int_0^t \left(\frac{1}{2}y^2\right)\Big|_{y=x-c(t-s)}^{y=x+c(t-s)} sds$$
$$= \int_0^t x(t-s)sds = \frac{1}{6}xt^3.$$

Or $u(x,t) = xt^3/6$.

Question 6

Solve $u_{tt} = c^2 u_{xx} + e^{ax}$, $u(x,0) = 0$, $u_t(x,0) = 0$.

Question 6 (Solution):

The solution formula for the wave equation with a source on the whole real axis is

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

For this problem, we have $\phi(x) = \psi(x) = 0$ and $f(x,t) = e^{ax}$. The solution is therefore

$$\begin{aligned}
u(x,t) &= \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} e^{ay} dyds = \frac{1}{2c}\int_0^t \left(\frac{1}{a}e^{ay}\right)\Big|_{y=x-c(t-s)}^{y=x+c(t-s)} sds \\
&= \frac{1}{2ca}\int_0^t \left[e^{a(x+ct-cs)} - e^{a(x-ct+cs)}\right] ds \\
&= \frac{1}{2ca}\left[\frac{1}{-ac}e^{a(x+ct-cs)} - \frac{1}{ac}e^{a(x-ct+cs)}\right]\Big|_{s=0}^{s=t} \\
&= \frac{1}{2ca}\left[\frac{1}{-ac}e^{a(x+ct-ct)} - \frac{1}{ac}e^{a(x-ct+ct)}\right] - \frac{1}{2ca}\left[\frac{1}{-ac}e^{a(x+ct)} - \frac{1}{ac}e^{a(x-ct)}\right] \\
&= \frac{1}{2c^2a^2}\left[-e^{ax} - e^{ax}\right] + \frac{1}{2c^2a^2}\left[e^{a(x+ct)} + e^{a(x-ct)}\right] \\
&= \frac{1}{2c^2a^2}\left[e^{a(x+ct)} + e^{a(x-ct)} - 2e^{ax}\right]
\end{aligned}$$

Or

$$u(x,t) = \frac{1}{2c^2a^2}\left[e^{a(x+ct)} + e^{a(x-ct)} - 2e^{ax}\right].$$

Question 7

Solve $u_{tt} = c^2 u_{xx} + \cos x,\ u(x,0) = \sin x,\ u_t(x,0) = 1 + x.$

Question 7 (Solution):

The solution formula for the wave equation with a source on the whole real axis is

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

For this problem, we have $\phi(x) = \sin x$, $\psi(x) = 1 + x$ and $f(x,t) = \cos x$. The solution is therefore

$$\begin{aligned}
u(x,t) &= \frac{1}{2}[\sin(x+ct) + \sin(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} (1+y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \cos y dyds \\
&= \sin x \cos(ct) + \frac{1}{2c}(2ct + 2xct) + \frac{1}{2c}\int_0^t (\sin[x + c(t-s)] - \sin[x - c(t-s)])\, ds \\
&= \sin x \cos(ct) + (t + xt) + \frac{1}{2c}\int_0^t 2\cos x \sin[c(t-s)]\, ds \\
&= \sin x \cos(ct) + (t + xt) + \frac{1}{c}\cos x \left(\frac{\cos[c(t-s)]}{c}\right)\Big|_{s=0}^{s=t} \\
&= \sin x \cos(ct) + (t + xt) + \frac{1}{c}\cos x \left(\frac{1}{c} - \frac{\cos(ct)}{c}\right) \\
&= \sin x \cos(ct) + t(1+x) + \frac{1}{c^2}\cos x - \frac{1}{c^2}\cos x \cos(ct).
\end{aligned}$$

Or

$$u(x,t) = \sin x \cos(ct) + t(1+x) + \frac{1}{c^2}\cos x - \frac{1}{c^2}\cos x \cos(ct).$$

Question 8

Consider the dynamics of a vibrating string with infinite length in the presence of an external force

$$\begin{cases} u_{tt} - u_{xx} = \delta(t-2)e^{-x}, & \text{for } -\infty < x < \infty, t > 0, \\ u(x,0) = \phi(x), \quad u_t(x,0) = 0 \end{cases}$$

where

$$\phi(x) = \begin{cases} 1, & \text{for } -1 < x < 1, \\ 0, & \text{else.} \end{cases}$$

Find the position of the point $x = 0$ on the string, $u(0,t)$, at any time t.

Question 8 (Solution):

For the inhomogeneous wave equation, we have the solution formula

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

Substituting $c = 1$, $f(x,t) = \delta(t-2)e^{-x}$, and $\psi(x) = 0$ into the above formula gives

$$\begin{aligned} u(x,t) &= \frac{1}{2}[\phi(x+t) + \phi(x-t)] + \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} \underline{\delta(s-2)e^{-y}}\, dyds \quad 0 < s < t \\ &= \frac{1}{2}[\phi(x+t) + \phi(x-t)] + \frac{1}{2}\int_0^t \delta(s-2)[e^{-x+(t-s)} - e^{-x-(t-s)}]ds. \end{aligned}$$

The solution has two parts depending on the value of t. That is,

$$u(x,t) = \begin{cases} \frac{1}{2}[\phi(x+t) + \phi(x-t)], & \text{for } t < 2, \\ \frac{1}{2}[\phi(x+t) + \phi(x-t)] + \frac{1}{2}[e^{-x+(t-2)} - e^{-x-(t-2)}], & \text{for } t \geq 2, \end{cases}$$

We are interested in $u(0,t)$; that is

$$\phi(t) = \begin{cases} 1, & -1 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$u(0,t) = \begin{cases} \frac{1}{2}[\phi(t) + \phi(-t)], & \text{for } t < 2, \\ \frac{1}{2}[\phi(t) + \phi(-t)] + \frac{1}{2}[e^{t-2} - e^{-(t-2)}], & \text{for } t \geq 2, \end{cases}$$

Question 8 (Solution):

We are interested in $u(0,t)$; that is

$$u(0,t) = \begin{cases} \frac{1}{2}[\phi(t) + \phi(-t)], & \text{for } t < 2, \\ \\ \frac{1}{2}[\phi(t) + \phi(-t)] + \frac{1}{2}[e^{t-2} - e^{-(t-2)}], & \text{for } t \geq 2, \end{cases}$$

Given that

$$\phi(x) = \begin{cases} 1, & \text{for } -1 < x < 1, \\ 0, & \text{else}, \end{cases}$$

we know

$$u(0,t) = \begin{cases} 1, & \text{for } 0 < t < 1, \\ 0, & \text{for } 1 \leq t < 2, \\ \frac{1}{2}[e^{t-2} - e^{-(t-2)}], & \text{for } t \geq 2. \end{cases}$$

Question 9

Consider the Neumann problem of a boundary source $g(t)$ on the half-line; that is,

$$
\begin{aligned}
&u_{tt} - c^2 u_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\
&u_x(0,t) = g(t) \quad \text{(boundary condition at } x = 0\text{)}, \\
&u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0\text{)}.
\end{aligned}
$$

Find its solution.

Question 9 (Solution):

We employ the subtraction method that we used for the heat equation to reduce the problem to one with zero Neuman data, and then use the reflection method to derive a solution formula for the reduced problem. Let

$$V(x,t) = u(x,t) - xg(t).$$

We can check that $V(x,t) = u(x,t) - xg(t)$ satisfies

$$\begin{cases} V_{tt} - c^2 V_{xx} = f(x,t) - xg''(t), \quad 0 < x < \infty, \quad t > 0, \\ V_x(0,t) = 0, \\ V(x,0) = \phi(x) - xg(0), \quad V_t(x,0) = \psi(x) - xg'(0). \end{cases}$$

This new problem has a homogeneous boundary condition to which we can apply the method of even extensions. For simplicity, let $F(x,t) = f(x,t) - xg''(t)$, $\Phi(x) = \phi(x) - xg(0)$, and $\Psi(x) = \psi(x) - xg'(0)$. Then,

$$\begin{aligned} &V_{tt} - c^2 V_{xx} = F(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0, \\ &V(x,0) = \Phi(x), \quad V_t(x,0) = \Psi(x). \end{aligned}$$

Take the even extensions:

$$\tilde{F}(x,t) = \begin{cases} F(x,t), & x \geq 0, \\ F(-x,t), & x < 0, \end{cases} \quad \tilde{\Phi}(x) = \begin{cases} \Phi(x), & x \geq 0, \\ \Phi(-x), & x < 0, \end{cases} \quad \tilde{\Psi}(x) = \begin{cases} \Psi(x), & x \geq 0, \\ \Psi(-x), & x < 0. \end{cases}$$

Question 9 (Solution):

Then we solve the extended wave equation on the whole real axis:

$$U_{tt} - c^2 U_{xx} = \tilde{F}(x,t), \quad -\infty < x < \infty, \quad t > 0,$$
$$U(x,0) = \tilde{\Phi}(x), \quad U_t(x,0) = \tilde{\Psi}(x), \quad -\infty < x < \infty,$$

Using the solution formula, we have the solution to the extended wave equation on the whole real axis; that is

$$U(x,t) = \frac{1}{2}[\tilde{\Phi}(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\Psi}(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \tilde{F}(y,s)dyds.$$

We only consider the solution $U(x,t) = V(x,t) = u(x,t) - xg(t)$ on the positive axis $x > 0$. Two different cases should be discussed.

Case 1: If $x > ct$ $(x > 0)$, then

$$V(x,t) = \frac{1}{2}[\Phi(x+ct) + \Phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \Psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(y,s)dyds.$$

Since $V(x,t) = u(x,t) - xg(t)$, we have

$$u(x,t) - xg(t) = \frac{1}{2}[\phi(x+ct) - xg(0) + \phi(x-ct) - xg(0)]$$
$$+ \frac{1}{2c}\int_{x-ct}^{x+ct} \left[\psi(y) - yg'(0)\right] dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \left[f(y,s) - yg''(s)\right] dyds.$$

Question 9 (Solution):

Since $V(x,t) = u(x,t) - xg(t)$, we have

$$u(x,t) - xg(t) = \frac{1}{2}[\phi(x+ct) - xg(0) + \phi(x-ct) - xg(0)]$$
$$+ \frac{1}{2c}\int_{x-ct}^{x+ct} \left[\psi(y) - yg'(0)\right] dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \left[f(y,s) - yg''(s)\right] dyds$$
$$= \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds$$
$$- xg(0) - xg'(0)t - \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} yg''(s)dyds$$
$$= \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds - xg(t).$$

Case 2: If $x < ct$ $(x > 0)$, then

$$V(x,t) = \frac{1}{2}[\Phi(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_0^{x+ct} \Psi(y)dy + \frac{1}{2c}\int_{x-ct}^0 \tilde{\Psi}(y)dy$$
$$+ \frac{1}{2c}\int_{t-\frac{x}{c}}^t \int_{x-c(t-s)}^{x+c(t-s)} F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}} \int_0^{x+c(t-s)} F(y,s)dyds$$
$$+ \frac{1}{2c}\int_0^{t-\frac{x}{c}} \int_{x-c(t-s)}^0 \tilde{F}(y,s)dyds$$

Question 9 (Solution):

Case 2: If $x < ct$ $(x > 0)$, then

$$V(x,t) = \frac{1}{2}[\Phi(x+ct)+\Phi(ct-x)] + \frac{1}{2c}\int_0^{x+ct}\Psi(y)dy + \frac{1}{2c}\int_{x-ct}^{0}\Psi(-y)dy$$
$$+\frac{1}{2c}\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)}F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)}F(y,s)dyds$$
$$+\frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_{x-c(t-s)}^{0}F(-y,s)dyds.$$

Make the change of variables in the blue integrals as y to $-y$, we get

$$V(x,t) = \frac{1}{2}[\Phi(x+ct)+\Phi(ct-x)] + \frac{1}{2c}\int_0^{x+ct}\Psi(y)dy + \frac{1}{2c}\int_0^{ct-x}\Psi(y)dy$$
$$+\frac{1}{2c}\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)}F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)}F(y,s)dyds$$
$$+\frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}F(y,s)dyds$$
$$= \frac{1}{2}[\Phi(x+ct)+\Phi(ct-x)] + \frac{1}{2c}\left\{\int_0^{x+ct}+\int_0^{ct-x}\right\}\Psi(y)dy$$
$$+\frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)}+\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)}+\int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\}F(y,s)dyds.$$

Question 9 (Solution):

Substituting into $F(x,t) = f(x,t) - xg''(t)$, $\Phi(x) = \phi(x) - xg(0)$, and $\Psi(x) = \psi(x) - xg'(0)$ gives

$$
\begin{aligned}
V(x,t) &= \frac{1}{2}[\phi(x+ct) - xg(0) + \phi(ct-x) - xg(0)] + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\left[\psi(y) - yg'(0)\right] dy \\
&\quad + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\}\left[f(y,s) - yg''(s)\right] dy \\
&= \frac{1}{2}[\phi(x+ct) + \phi(ct-x)] - xg(0) + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\psi(y)dy \\
&\quad - \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\} yg'(0)dy \\
&\quad + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} f(y,s)dyds \\
&\quad - \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} yg''(s)dyds
\end{aligned}
$$

Question 9 (Solution):

It can be further simplied as

$$
\begin{aligned}
V(x,t) = {} & \frac{1}{2}[\phi(x+ct)+\phi(ct-x)] + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\psi(y)dy \\
& + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} f(y,s)dyds \\
& -xg(0) - \frac{1}{2c}(x^2+c^2t^2)g'(0) + \frac{x^2}{c}g'\left(t-\frac{x}{c}\right) - xg(t) + xg\left(t-\frac{x}{c}\right) \\
& -\frac{x^2}{2c}\left[g'\left(t-\frac{x}{c}\right) - g(0)\right] \\
& -\frac{x^2}{2c}g'\left(t-\frac{x}{c}\right) + \frac{ct^2}{2}g'(0) - xg\left(t-\frac{x}{c}\right) + ctg(0) - c\int_0^{t-\frac{x}{c}} g(s)ds
\end{aligned}
$$

Further,

$$
\begin{aligned}
V(x,t) = {} & \frac{1}{2}[\phi(x+ct)+\phi(ct-x)] + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\psi(y)dy \\
& + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} f(y,s)dyds \\
& -xg(0) - xg(t) + ctg(0) - c\int_0^{t-\frac{x}{c}} g(s)ds
\end{aligned}
$$

Question 9 (Solution):

Since $V(x,t) = u(x,t) - xg(t)$, when $x < ct$ $(x > 0)$ we have

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(ct-x)] + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\psi(y)dy$$
$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} f(y,s)dyds$$
$$-xg(0)+ctg(0) - c\int_0^{t-\frac{x}{c}} g(s)ds$$

Therefore, the solution to the Neumann problem of a boundary source $g(t)$ on the half-line $(x > 0)$ is when $x > ct$ $t < x/c$

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct}\psi(y)dy + \frac{1}{2c}\int_0^t\int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds.$$

when $x < ct$

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(ct-x)] + \frac{1}{2c}\left\{\int_0^{x+ct} + \int_0^{ct-x}\right\}\psi(y)dy$$

$t > x/c$

$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_0^{-x+c(t-s)}\right\} f(y,s)dyds$$
$$-xg(0)+ctg(0) - c\int_0^{t-\frac{x}{c}} g(s)ds.$$

MH4110 PDE

Tutorial 08

Question 1

Solve $u_t = ku_{xx}$; $u(x,0) = 0$; $u(0,t) = 1$ on the half-line $0 < x < \infty$.

Question 1

Solve $u_t = ku_{xx}$; $u(x,0) = 0$; $u(0,t) = 1$ on the half-line $0 < x < \infty$.

We employ the subtraction method to reduce the problem to one with zero Dirichlet data, and then use the reflection method to derive a solution formula for the reduced problem. Let

$$U(x,t) = u(x,t) - 1.$$

We can check that $U(x,t) = u(x,t) - 1$ satisfies

$$\begin{aligned} &U_t - kU_{xx} = 0, \quad 0 < x < \infty, \quad t > 0, \\ &U(0,t) = 0, \\ &U(x,0) = -1. \end{aligned}$$

This new problem has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions. From the lecture, the solution can be written as

$$U(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) - \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.$$

Here $\phi(y) = -1$. Thus,

$$\begin{aligned} U(x,t) &= \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[-\exp\Big(-\frac{(x-y)^2}{4kt}\Big) + \exp\Big(-\frac{(x+y)^2}{4kt}\Big)dy \\ &= -\frac{1}{\sqrt{4\pi kt}} \int_0^\infty \exp\Big(-\frac{(x-y)^2}{4kt}\Big)dy + \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \exp\Big(-\frac{(x+y)^2}{4kt}\Big)dy. \end{aligned}$$

Question 1

Solve $u_t = ku_{xx}$; $u(x,0) = 0$; $u(0,t) = 1$ on the half-line $0 < x < \infty$.

$$U(x,t) = -\frac{1}{\sqrt{4\pi kt}}\int_0^\infty \exp\Big(-\frac{(x-y)^2}{4kt}\Big)dy + \frac{1}{\sqrt{4\pi kt}}\int_0^\infty \exp\Big(-\frac{(x+y)^2}{4kt}\Big)dy.$$

Let $p = \frac{y-x}{\sqrt{4kt}}$ and $q = \frac{y+x}{\sqrt{4kt}}$, then

$$\begin{aligned}
U(x,t) &= -\frac{1}{\sqrt{\pi}}\int_{\frac{-x}{\sqrt{4kt}}}^\infty \exp\Big(-p^2\Big)dp + \frac{1}{\sqrt{\pi}}\int_{\frac{x}{\sqrt{4kt}}}^\infty \exp\Big(-q^2\Big)dq \\
&= \frac{1}{\sqrt{\pi}}\int_\infty^{\frac{-x}{\sqrt{4kt}}} \exp\Big(-p^2\Big)dp + \frac{1}{\sqrt{\pi}}\int_{\frac{x}{\sqrt{4kt}}}^\infty \exp\Big(-q^2\Big)dq \\
&= \frac{1}{\sqrt{\pi}}\int_{\frac{x}{\sqrt{4kt}}}^{\frac{-x}{\sqrt{4kt}}} \exp\Big(-p^2\Big)dp \\
&= -Erf\left(\frac{x}{\sqrt{4kt}}\right).
\end{aligned}$$

The solution of the original problem is therefore

$$u(x,t) = 1 - Erf\left(\frac{x}{\sqrt{4kt}}\right).$$

Question 2

Solve $u_{tt} = 4u_{xx}$ for $0 < x < \infty$, $u(0,t) = 0$, $u(x,0) \equiv 1$, $u_t(x,0) \equiv 0$ using the reflection method. This solution has a singularity; find its location.

Question 2

Solve $u_{tt} = 4u_{xx}$ for $0 < x < \infty$, $u(0,t) = 0$, $u(x,0) \equiv 1$, $u_t(x,0) \equiv 0$ using the reflection method. This solution has a singularity; find its location.

This is an initial boundary value problem with zero Dirichlet data. Its solution is

$$u(x,t) = \begin{cases} \frac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\ \frac{1}{2}\left[\phi(x+ct) - \phi(ct-x)\right] + \frac{1}{2c}\int_{ct-x}^{ct+x} \psi(s)ds, & 0 < x < ct. \end{cases}$$

In this problem we have $c = 2$, $\phi(x) = 1$, and $\psi(x) = 0$. Thus, the solution is

$$u(x,t) = \begin{cases} \frac{1}{2}\left[\phi(x+2t) + \phi(x-2t)\right] + \frac{1}{4}\int_{x-2t}^{x+2t} \psi(s)ds, & x > 2t, \\ \frac{1}{2}\left[\phi(x+2t) - \phi(2t-x)\right] + \frac{1}{4}\int_{2t-x}^{2t+x} \psi(s)ds, & 0 < x < 2t. \end{cases}$$

It can be expressed as

$$u(x,t) = \begin{cases} 1, & x > 2t, \\ 0, & x < 2t. \end{cases}$$

$t < \frac{x}{2}$

$t > x/2$

The solution has a jump discontinuity at $x = 2t$. To the right of $(x, x/2)$, the displacement is 1; while the displacement is 0 to the left due to the cancellation from the boundary point at $x = 0$.

Question 3

Solve $u_{tt} = c^2 u_{xx}$ for $0 < x < \infty$, $u(0,t) = t^2$, $u(x,0) = x$, $u_t(x,0) = 0$.

Question 3

Solve $u_{tt} = c^2 u_{xx}$ for $0 < x < \infty$, $u(0,t) = t^2$, $u(x,0) = x$, $u_t(x,0) = 0$.

A boundary source $h(t) = t^2$ exists at the boundary $x = 0$. From the lecture, we know that the solution to the Dirichlet problem of a boundary source $h(t)$ on the half-line ($x > 0$) is when $x > ct$

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds;$$

when $x < ct$

$$u(x,t) = h(t - \frac{x}{c}) + \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds.$$

Given that $\phi(x) = x$, $\psi(x) = 0$, and $f(x,t) = 0$, the solution is therefore

$$u(x,t) = \begin{cases} \frac{1}{2}[(x+ct) + (x-ct)] = x, & \text{when } x > ct, \\ (t - \frac{x}{c})^2 + \frac{1}{2}[(x+ct) - (ct-x)] = (t - \frac{x}{c})^2 + x, & \text{when } x < ct. \end{cases}$$

Question 4

Consider waves in a resistant medium that satisfy the problem

$$\begin{cases} u_{tt} = c^2 u_{xx} - r u_t, & \text{for } 0 < x < l, \\ u(0,t) = u(l,t) = 0, \\ u(x,0) = \phi(x), u_t(x,0) = \psi(x), \end{cases}$$

where r is a constant, $0 < r < 2\pi c/l$. Write down the series expansion of the solution.

Question 4

Consider waves in a resistant medium that satisfy the problem

$$\begin{cases} u_{tt} = c^2 u_{xx} - r u_t, & \text{for } 0 < x < l, \\ u(0,t) = u(l,t) = 0, \\ u(x,0) = \phi(x), u_t(x,0) = \psi(x), \end{cases}$$

where r is a constant, $0 < r < 2\pi c/l$. Write down the series expansion of the solution.

Look for the solution: $u(x,t) = X(x)T(t)$. Then we have

$$XT'' = c^2 X'' T - rXT'.$$

Dividing both sides by $-c^2 XT$, gives

$$-\frac{T'' + rT'}{c^2 T} = -\frac{X''}{X} := \lambda.$$

This yields the eigen-problem:

$$X'' + \lambda X = 0, \quad X(0) = X(l) = 0,$$

and the ODE for T :

$$T'' + rT' + c^2 \lambda T = 0.$$

Question 4

This yields the eigen-problem:

$$X'' + \lambda X = 0, \quad X(0) = X(l) = 0,$$

and the ODE for T :

$$T'' + rT' + c^2\lambda T = 0.$$

$$\frac{-r \pm \sqrt{c^2\lambda - r}\, i}{2}$$

The eigenvalues and eigenfunctions are

$$\lambda_n = \frac{n^2\pi^2}{l^2}, \quad X_n(x) = \sin\left(\frac{n\pi x}{l}\right), \quad n = 1, 2, \cdots.$$

We turn to the ODE for T :

$$T_n(t) = e^{-rt/2}(A_n \cos\omega_n t + B_n \sin\omega_n t), \quad \omega_n = \sqrt{c^2 n^2\pi^2/l^2 - r^2/4}. \geq 0$$

The solution formula is

$$u(x,t) = \sum_{n=1}^{\infty} e^{-rt/2}(A_n \cos\omega_n t + B_n \sin\omega_n t)\sin(n\pi x/l),$$

for all n

where A_n and B_n are determined by the initial conditions.

Question 5

Consider waves in a resistant medium that satisfy the problem

$$\begin{cases} u_{tt} = c^2 u_{xx} - r u_t, & \text{for } 0 < x < l, \\ u(0,t) = u(l,t) = 0, \\ u(x,0) = \phi(x), u_t(x,0) = \psi(x), \end{cases}$$

where r is a constant, $4\pi c/l < r < 6\pi c/l$. Write down the series expansion of the solution.

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Look for the solution: $u(x,t) = X(x)T(t)$. Then we have

$$XT'' = c^2 X'' T - rXT'.$$

Dividing both sides by $-c^2 XT$, gives

$$-\frac{T'' + rT'}{c^2 T} = -\frac{X''}{X} := \lambda.$$

This yields the eigen-problem:

$$X'' + \lambda X = 0, \quad X(0) = X(l) = 0,$$

and the ODE for T :

$$T'' + rT' + c^2 \lambda T = 0.$$

$r^2 - 4c^2\lambda < 0$ for λ_n, $n > 2$

$r^2 - 4c^2\lambda > 0$ for λ_1, λ_2

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We turn to the ODE for T. If $4\pi c/l < r < 6\pi c/l$, when $n = 1, 2$

$$T_1(t) = e^{-rt/2}\left(A_1 e^{t\sqrt{r^2/4 - c^2\pi^2/l^2}} + B_1 e^{-t\sqrt{r^2/4 - c^2\pi^2/l^2}}\right).$$

$$T_2(t) = e^{-rt/2}\left(A_2 e^{t\sqrt{r^2/4 - 4c^2\pi^2/l^2}} + B_2 e^{-t\sqrt{r^2/4 - 4c^2\pi^2/l^2}}\right).$$

When $n > 2$,

$$T_n(t) = e^{-rt/2}\left(A_n \cos\omega_n t + B_n \sin\omega_n t\right), \quad \omega_n = \sqrt{c^2 n^2\pi^2/l^2 - r^2/4}.$$

The solution formula is

$$\begin{aligned} u(x,t) = & \underbrace{e^{-rt/2}\left(A_1 e^{t\sqrt{r^2/4 - c^2\pi^2/l^2}} + B_1 e^{-t\sqrt{r^2/4 - c^2\pi^2/l^2}}\right)}_{T_1} \underbrace{\sin(\pi x/l)}_{X_1} \\ & + \underbrace{e^{-rt/2}\left(A_2 e^{t\sqrt{r^2/4 - 4c^2\pi^2/l^2}} + B_2 e^{-t\sqrt{r^2/4 - 4c^2\pi^2/l^2}}\right)}_{T_2} \underbrace{\sin(2\pi x/l)}_{X_2} \\ & + \sum_{n=3}^{\infty} \underbrace{e^{-rt/2}\left(A_n \cos\omega_n t + B_n \sin\omega_n t\right)}_{T_n} \underbrace{\sin(n\pi x/l)}_{X_n \ (n>2)}, \end{aligned}$$

where A_n and B_n are determined by the initial conditions.